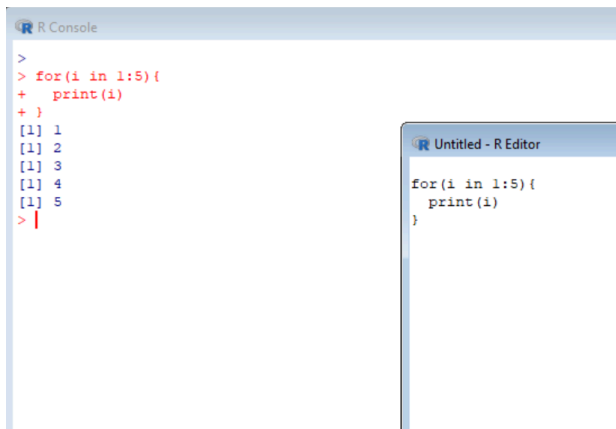


FOR LOOP

Q1. Print numbers 1 to 5

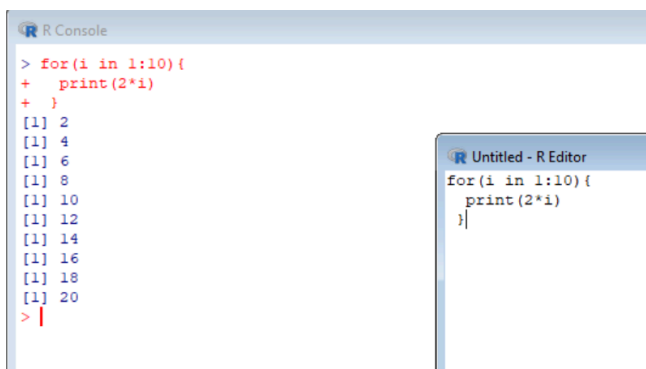
```
for(i in 1:5){  
  print(i)  
}
```



The screenshot shows two windows from the R environment. The 'R Console' window on the left displays the execution of the code: the prompt '>' is followed by the code 'for(i in 1:5){', then '+ print(i)', and '+ }'. Below this, the output is shown as '[1] 1', '[1] 2', '[1] 3', '[1] 4', and '[1] 5', each on a new line. The prompt '>' is at the bottom with a red cursor. The 'Untitled - R Editor' window on the right shows the source code: 'for(i in 1:5){', ' print(i)', and '}'.

Q2. Print multiplication table of 2

```
for(i in 1:10){  
  print(2*i)  
}
```

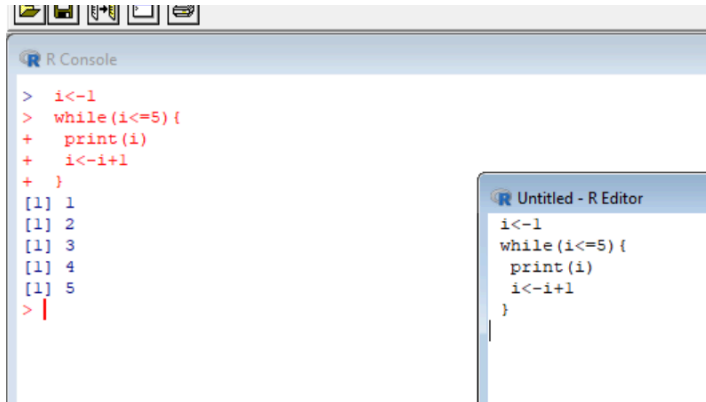


The screenshot shows two windows from the R environment. The 'R Console' window on the left displays the execution of the code: the prompt '>' is followed by the code 'for(i in 1:10){', then '+ print(2*i)', and '+ }'. Below this, the output is shown as '[1] 2', '[1] 4', '[1] 6', '[1] 8', '[1] 10', '[1] 12', '[1] 14', '[1] 16', '[1] 18', and '[1] 20', each on a new line. The prompt '>' is at the bottom with a red cursor. The 'Untitled - R Editor' window on the right shows the source code: 'for(i in 1:10){', ' print(2*i)', and '}'.

WHILE LOOP

Q1. Print numbers 1 to 5

```
i<-1
while(i<=5){
  print(i)
  i<-i+1
}
```



The screenshot shows the R environment. The R Console on the left displays the execution of the while loop code, showing the output [1] 1, [1] 2, [1] 3, [1] 4, and [1] 5. The R Editor on the right shows the source code for the while loop.

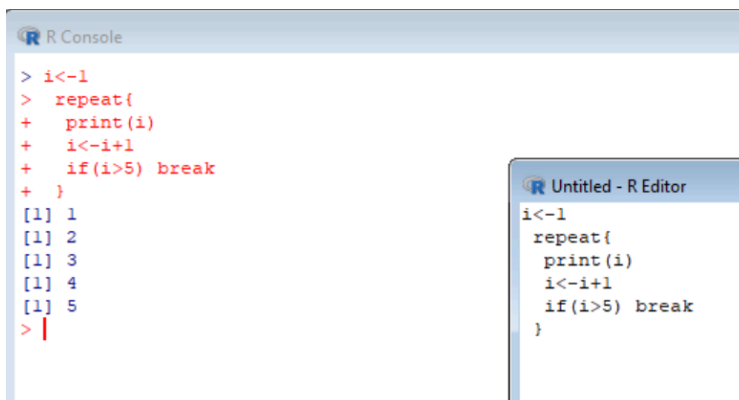
```
R Console
> i<-1
> while(i<=5){
+   print(i)
+   i<-i+1
+ }
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
> |

Untitled - R Editor
i<-1
while(i<=5){
  print(i)
  i<-i+1
}
```

REPEAT LOOP

Q1. Print 1 to 5 using repeat

```
i<-1
repeat{
  print(i)
  i<-i+1
  if(i>5) break
}
```



The screenshot shows the R environment. The R Console on the left displays the execution of the repeat loop code, showing the output [1] 1, [1] 2, [1] 3, [1] 4, and [1] 5. The R Editor on the right shows the source code for the repeat loop.

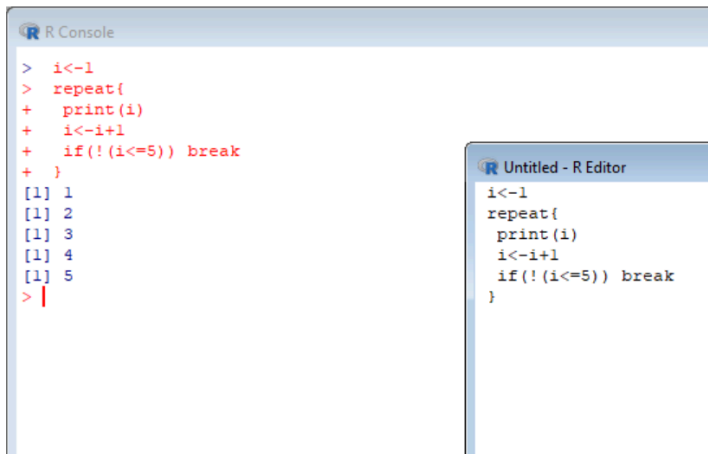
```
R Console
> i<-1
> repeat{
+   print(i)
+   i<-i+1
+   if(i>5) break
+ }
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
> |

Untitled - R Editor
i<-1
repeat{
  print(i)
  i<-i+1
  if(i>5) break
}
```

DO WHILE (simulated)

Q1. Execute at least once

```
i<-1
repeat{
  print(i)
  i<-i+1
  if(!(i<=5)) break
}
```



The screenshot shows two windows from the R environment. The 'R Console' window on the left displays the execution of the code: the variable `i` is initialized to 1, and the `repeat` loop prints the values 1, 2, 3, 4, and 5 before breaking. The 'Untitled - R Editor' window on the right shows the source code for the same script.

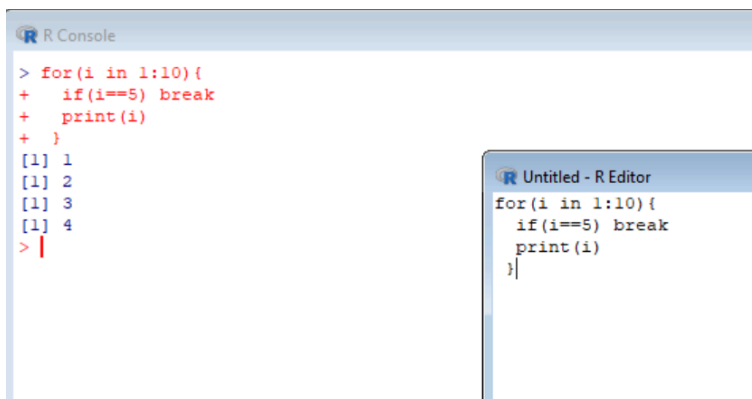
```
R Console
> i<-1
> repeat{
+   print(i)
+   i<-i+1
+   if(!(i<=5)) break
+ }
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
> |

Untitled - R Editor
i<-1
repeat{
  print(i)
  i<-i+1
  if(!(i<=5)) break
}
```

BREAK

Q1. Stop at 5

```
for(i in 1:10){
  if(i==5) break
  print(i)
}
```



The screenshot shows two windows from the R environment. The 'R Console' window on the left displays the execution of the code: the `for` loop iterates from 1 to 10, but the `break` statement stops the loop after printing the value 4. The 'Untitled - R Editor' window on the right shows the source code for the same script.

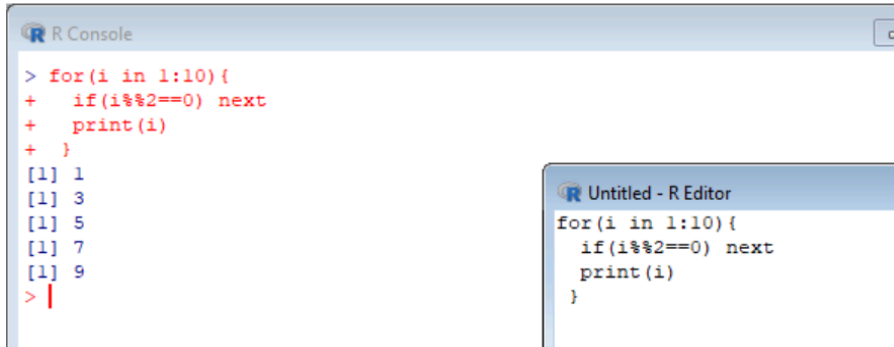
```
R Console
> for(i in 1:10){
+   if(i==5) break
+   print(i)
+ }
[1] 1
[1] 2
[1] 3
[1] 4
> |

Untitled - R Editor
for(i in 1:10){
  if(i==5) break
  print(i)
}
```

NEXT

Q1. Skip even numbers

```
for(i in 1:10){  
  if(i%%2==0) next  
  print(i)  
}
```



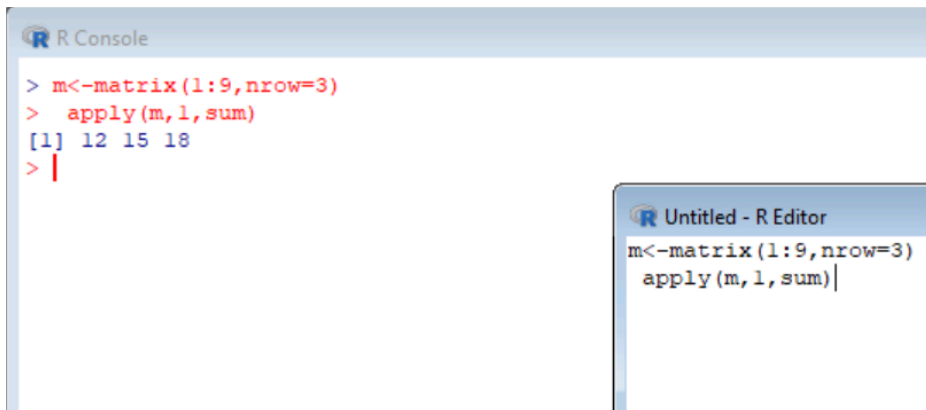
The screenshot shows an R Console window on the left and an R Editor window on the right. The R Console displays the execution of the provided R code, showing the output of the loop: 1, 3, 5, 7, and 9. The R Editor shows the source code of the script being executed.

```
R Console  
> for(i in 1:10){  
+   if(i%%2==0) next  
+   print(i)  
+ }  
[1] 1  
[1] 3  
[1] 5  
[1] 7  
[1] 9  
> |  
  
Untitled - R Editor  
for(i in 1:10){  
  if(i%%2==0) next  
  print(i)  
}
```

APPLY

Q1. Row sums of matrix

```
m<-matrix(1:9,nrow=3)  
apply(m,1,sum)
```



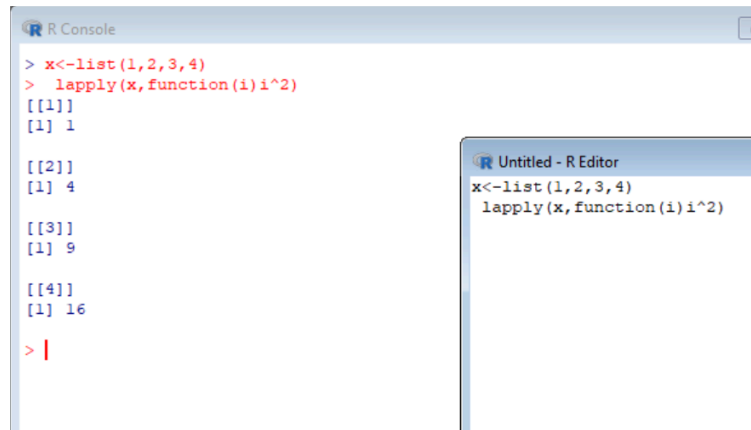
The screenshot shows an R Console window on the left and an R Editor window on the right. The R Console displays the execution of the provided R code, showing the output of the apply function: 12, 15, and 18. The R Editor shows the source code of the script being executed.

```
R Console  
> m<-matrix(1:9,nrow=3)  
> apply(m,1,sum)  
[1] 12 15 18  
> |  
  
Untitled - R Editor  
m<-matrix(1:9,nrow=3)  
apply(m,1,sum)|
```

LAPPLY

Q1. Square list elements

```
x<-list(1,2,3,4)
lapply(x,function(i) i^2)
```



The image shows two windows from an R environment. The 'R Console' window on the left displays the execution of the code: `x<-list(1,2,3,4)` and `lapply(x,function(i) i^2)`. The output is a list with four elements, each shown as `[[i]]` followed by `[1] value`, where the values are the squares of 1, 2, 3, and 4 (1, 4, 9, 16). The 'Untitled - R Editor' window on the right shows the same code being executed.

```
R Console
> x<-list(1,2,3,4)
> lapply(x,function(i) i^2)
[[1]]
[1] 1

[[2]]
[1] 4

[[3]]
[1] 9

[[4]]
[1] 16

> |

Untitled - R Editor
x<-list(1,2,3,4)
lapply(x,function(i) i^2)
```